Inbrev

Hack Apertus 2026 · Track 2B · Team Apertool

The practice inbox that never leaves the practice.

Inbrev sorts the inbox of a Swiss GP practice by handling path and deadline, with Apertus, built to run on a box in the practice. Every field it extracts quotes its source.

▶ Watch the 2-min demo

Code on GitHub [TBD link]

solo builder · fictional data only

THE PROBLEM

Before the first patient, there is the inbox.

Lab reports, refill requests, insurer forms, referrals and faxes in German, French and Italian all land in one queue. A medical practice assistant (MPA) reads them one by one.

9.5 h / week

of administration for Swiss GPs and paediatricians in 2025, up from 8.6 h in 2020.

Zeller, Fischer, Hernandez, *Praxis* 2026 · 5th Workforce Study, n = 1,776

54 min / day

that practice-based doctors spend on requests from authorities and insurers alone.

gfs.bern for FMH, 2025 · N = 1,532 doctors

Art. 321

Swiss Criminal Code: professional secrecy binds doctors **and their assistants**. Every message in that inbox is covered.

SCC, SR 311.0 · health data are sensitive personal data under FADP Art. 5

HOW IT WORKS · PURPOSEFUL USE OF AI

Sorted by who handles it and by when. Every field quotes its source.

01 · READ

Every channel

E-mail, HIN mail, web forms and scanned faxes (OCR). German, French, Italian and English.

02 · ROUTE

9 handling paths

Report, appointment, refill, insurer, referral, certificate, billing, marketing, other. Plus an administrative deadline and who takes it: MPA, doctor or admin.

03 · PROVE

Quote or drop

Every field carries the exact words it came from. Code checks that the quote is in the original. If Inbrev is unsure, **a human decides**.

Clinical content A message that may contain symptoms is excluded from automation and goes to a person, like any message Inbrev is unsure about. Inbrev does not judge it and does not rank it.

Apertus reads free text in four languages. Everything code can check (AHV numbers, dates, insurer names, quotes) is checked by code. Small local models do best on narrowly cut tasks (Xu et al., *JMIR* 2026, CHUV).

DEMO

One click. Sorted by path and deadline; what needs a person goes to a person.

Inbox · sorted		8 of 30 shown
YOU DECIDE	**Peter Huber** Two topics in one message – not sure	MPA
CLINICAL CONTENT	**Heidi Zimmermann** Rezept – und seit heute Druck auf der Brust	person
TODAY	**Labor Rheintal AG** Report received · Lukas Weber	doctor
THIS WEEK	**Fax · KV Beispiel AG (fict.)** Insurer · Kostengutsprache Physio	admin
THIS WEEK	**Marc Rochat · FR** Refill · Renouvellement ordonnance	MPA
THIS WEEK	**Tobias Frei · web form** Certificate · Arbeitsunfähigkeit	MPA
NO DEADLINE	**Dr. P. Bernasconi · IT** Referral · Invio paziente	MPA

Fax · OCR · German · route **insurer** → admin · deadline **08.10.2026**

```
KOSTENGUTSPRACHE – RÜCKFRAGE
Versicherte Person: Anna Meier, 22.08.1979,
AHV 756.1234.9876.14
Wir benötigen bis 08.10.2026 die ärztliche
Verordnung für 9 Sitzungen Physiotherapie. Referenz
KG-2026-55813.
```

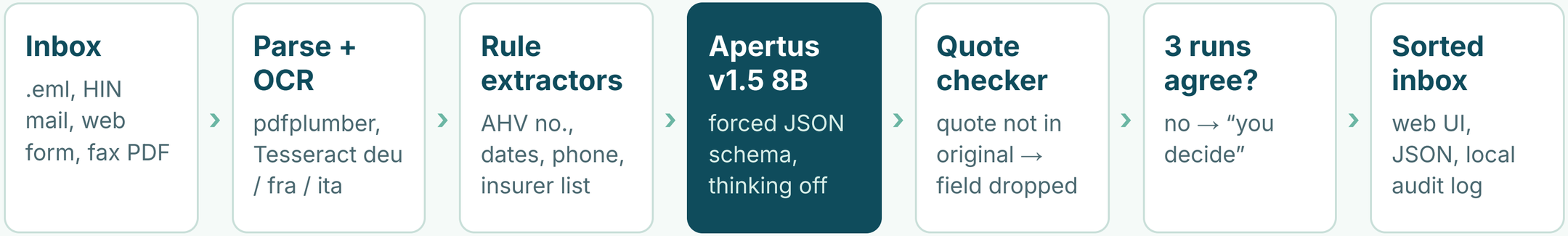
Patient	**Anna Meier** “Anna Meier”
Date of birth	**1979-08-22** “22.08.1979”
AHV no.	**756.1234.9876.14** “format check: code”
Deadline	**2026-10-08** “bis 08.10.2026”
Insurer ref.	**KG-2026-55813** “Referenz KG-2026-55813”

Mock-up with messages from the fictional seed set; no real patient data anywhere. [TBD: replace with screenshot of the real app]

ARCHITECTURE · SOVEREIGN DEPLOYABILITY

Pull the cable. Inbrev keeps sorting.

INSIDE THE PRACTICE · NO OUTBOUND CONNECTION AT RUN TIME



On-prem box	Mini PC with GPU in a group practice
Air-gapped	Model host offline; a mail gateway hands messages in
Swiss cloud	Option for single practices, Swiss hosting
CPU only	GGUF Q4 for judges without a GPU

```
$ docker run --rm --network none \
    inbrev python -m pytest -q
18 passed
Full triage on local GGUF weights: not run yet

# downloaded at build time only:
# weights (SHA-256), images, OCR data
```

EVALUATION · TECHNICAL RIGOUR

Measured: what leaves automation, and what still runs by itself.

180 synthetic messages, labels not yet reviewed by a person. Target 0.95 for clinical exclusion recall and “today” recall; 8B does not reach it yet (13 of 16, 95 % CI 0.57–0.93).

Setup	Macro-F1 9 paths	Clinical exclusion recall	Recall “today”	Runs automatically
Keyword rules + lexical net	0.76	0.44 (7/16)	0.76	100 %
TF-IDF + log. regression	0.82	0.13 (2/16)	0.66	100 %
Apertus 8B, one answer, no checks	0.84	0.56 (9/16)	0.70	100 %
Inbrev with Apertus 70B	0.89	0.94 (15/16)	0.94	94 %
Inbrev (8B + checks + abstain)	**0.85**	**0.81 (13/16)**	**0.86**	**96 %**

Honest read: the keyword rules already route 79 % correctly and run without a model; on 30 extra Swiss-style messages they even route better than 8B (25 vs 23 of 30). The model earns its place on clinical exclusion (7 → 13 of 16) and on fields (F1 0.73 with 8B, 0.83 with 70B). About half of the flags are false alarms from the lexical net. Not measured: other open models, fax OCR.

Test set v0: 180 fictional messages

- Labels are fixed before any text is written.
- Text by two other model families (Claude, GPT-5.5), 90 each. Apertus never writes or labels test data.
- Fictional people; AHV numbers with an invalid check digit.
- Planned: handwritten messages, scanned faxes and a blind relabelling by a person (κ)
- Open data, CDLA-Permissive-2.0.

VALUE, COST & SCALABILITY

Swiss francs per 1,000 messages, on hardware the practice owns.

ON-PREM GPU

Apertus 8B · RTX 5070 Ti

CHF TBD

per 1,000 messages (power + depreciation)

Latency: TBD s / message

Energy: TBD Wh / message

CPU ONLY

GGUF Q4, no GPU

CHF TBD

per 1,000 messages

Latency: TBD s / message

Fine for an overnight batch? TBD

REFERENCE

Swiss cloud API

≈ CHF 3.35

per 1,000 requests of 3,000 input + 500 output tokens

Inbrev tokens / message (8B, measured): 1,398 in, 644 out

3 samples in one request

From CHF 0.00335 per request, blog estimate (Pupel, devs-group, 2026).

ASSUMPTION

What it buys: 60 messages a day × 1 minute each × CHF 50/h MPA time ≈ CHF 50 a day. To be checked in interviews with practices.

INTENDED PURPOSE · IMPLEMENTATION FEASIBILITY

Administrative by design. Inbrev routes, it never triages.

Inbrev does

- ✓ Sort by handling path and administrative deadline
- ✓ Extract patient and case fields, each with its quote
- ✓ Hand clinical content to a person, unassessed and unranked
- ✓ Say “you decide” when it is unsure

Inbrev never does

- ✗ Rank messages by clinical severity
- ✗ Assess symptoms or suggest care
- ✗ Make a decision a person has not seen
- ✗ Send patient data out of the practice

Why the line sits here: MDCG 2019-11 Rev.1 (2025) treats software that automatically reprioritises by clinical criticality based on patient data as medical device software; software that only stores, communicates or searches is not. Switzerland classifies via MepV Art. 15 → EU MDR Annex VIII. Privacy: FADP Art. 5, 7 and 22, and SCC Art. 321.

BUSINESS MODEL & ROADMAP

A module for the software practices already use.

Who uses it	The MPA, first thing in the morning
Who pays	The practice owner or the doctors' network
Price	CHF 150–250 per practice and month HYPOTHESIS
Channel	As a module inside HIN mail or practice software, not one more window
Open core	Hackathon code Apache-2.0. Revenue from operation, connectors, support and updates
Later	Germany and Austria: same logic, same secrecy duty (§ 203 StGB)

OCT 2026 · HACK APERTUS
Open gold set, full evaluation, offline proof, CHF per 1,000 messages

NOV – DEC 2026
5–10 interviews with MPAs and practice owners: real volumes, categories, fax share

Q1 2027
Pilot in one group practice on its own box; mail and practice-software connectors

Q2 2027
Written intended purpose, impact assessment, classification review before first sale

Inbrev

Team Apertool · Niklas Wichter, solo builder

The practice inbox that never leaves the practice.

We are looking for one Swiss group practice to pilot Inbrev on its own hardware in Q1 2027.

Contact: niklaswichter@gmail.com

Demo ▶ 2-min video **Code** `[TBD GitHub link]` **Data** `[TBD Hugging Face link]`